

For each set of vectors, T , determine whether or not they are (i) linearly independent or dependent, (ii) span $\mathbb{R}^n$ for the appropriate n , (iii) form a basis for $\mathbb{R}^n$ for the appropriate n .

$$(1) T_1 = \left\{ \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 9 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 4 & 9 & -1 \end{bmatrix} \quad \rightarrow \quad R = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(2) T_2 = \left\{ \mathbf{v}_1 = \begin{bmatrix} 12 \\ -9 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -10 \\ -1 \\ 3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ -5 \\ 12 \end{bmatrix} \right\}$$

$$A = \begin{bmatrix} 12 & -10 & 5 \\ -9 & -1 & -5 \\ 1 & 3 & 12 \end{bmatrix} \quad \rightarrow \quad R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(3) T_3 = \left\{ \mathbf{v}_1 = \begin{bmatrix} 12 \\ -9 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -10 \\ -1 \\ 3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ -5 \\ 12 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 7 \\ -15 \\ 16 \end{bmatrix} \right\}$$

$$(4) \quad T_4 = \left\{ \mathbf{v}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 5 \\ -1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 6 \end{bmatrix} \right\}$$

$$A = \begin{bmatrix} 2 & -1 & 5 & 2 \\ 2 & 1 & -1 & 6 \end{bmatrix} \quad \rightarrow \quad R = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -3 & 2 \end{bmatrix}$$

$$(5) \quad T_5 = \left\{ \mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 5 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ -1 \\ 6 \end{bmatrix} \right\}$$

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \\ 5 & -1 \\ 2 & 6 \end{bmatrix} \quad \rightarrow \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$